

Final Submission Document

Cover Page

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- Course: SFIA Level 3 Integrated Assessment
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Executive Summary

This submission presents the completed Student Support Insights Tool prototype. The final system helps programme teams upload or capture learner support data, validate it, view dashboard trends, identify learners who may need earlier support, and generate practical recommendations for human-reviewed intervention planning. Synthetic data was used throughout to reduce privacy risk while still demonstrating realistic system behaviour.

The final version also includes settings-driven display preferences, ethics confirmation before sensitive actions, confirmation on every data upload, confirmation before clearing report state, and refreshed screenshot and build evidence for submission review.

1. Problem Statement And Context

Digital skills programmes often collect useful learner information, but the data is not always analysed early enough or consistently enough to support timely intervention. This can delay support and reduce consistency in staff decision-making. This project addresses the following core question:

How can a programme team use learner data responsibly to identify support needs, prioritise support actions, and improve learner success?

2. Stakeholder And User Needs

Primary stakeholders include learners, programme staff, leadership, IT support, and mentors. Learners require timely and fair support. Staff require clear evidence for prioritisation. Leadership requires performance visibility for planning and resource allocation. Supporting models:

- Value model: docs/BusinessModels/value_model_diagram.md

- Completed peer/stakeholder review record:
docs/Documentation/completed_peer_stakeholder_review.md
- Assessor walkthrough review:
docs/Documentation/assessor_walkthrough_review.md

3. Current Process And Improved Process

The as-is process is mostly manual and depends on spreadsheet review. The to-be process is more organised: capture or upload learner data, validate it, generate dashboard analytics, flag possible support risk, produce recommendations, and let staff review the result before taking action.

- As-is model: docs/ProcessModels/as_is_process_template.md
- To-be model: docs/ProcessModels/to_be_process_template.md

4. Data Description And Preparation

Dataset: 30 synthetic learner records with fields for access, confidence, support need, and attendance risk. Data quality controls implemented:

- Missing value checks
- Controlled-value validation for age band, device access, internet access, and attendance risk
- Duplicate learner ID detection
- Confidence range checks (1 to 5)
- Quote-aware CSV parsing for notes containing commas
- Row-level validation errors with line numbers for rejected uploads
- Inline manual-form validation support

Supporting evidence:

- Dataset: data/Dataset/synthetic_learners.csv
- Data dictionary: data/Dataset/data_dictionary.md

5. Analytics Findings And Visualizations

The final prototype exceeds the minimum visualization requirement with five implemented charts:

- Risk distribution across Low, Medium, High, and Critical categories
- Support need distribution
- Device access distribution
- Internet access by province

- Confidence comparison across digital, programming, and AI familiarity scores

Key findings include concentration of high-risk learners, internet and device constraints, uneven confidence across digital capability areas, and recurring support needs that can be prioritised for intervention planning.

- Findings: docs/FinalReport/analytics_findings.md

6. Prototype Description

Implemented features:

- CSV upload and manual entry
- Inline and row-level validation feedback
- Dashboard with KPI cards and 5 charts
- Searchable and filterable high-risk learner table with score, priority, and reason codes
- Recommendation output (3 practical recommendations)
- Ethics/privacy notice with human review checklist
- Help and Settings modal workflow
- Settings persistence and configurable dashboard display controls
- Upload confirmation on every file upload
- New Report confirmation before clearing loaded learner data
- CSV export

Technical implementation:

- Backend: ASP.NET Core (.NET 10)
- Frontend: React + TypeScript + Vite + Recharts
- Main run guide: README.md

7. AI, Data Ethics, And Privacy Review

The ethics section was strengthened because the tool works with learner support indicators such as access, confidence, attendance risk and support needs. Even without names, this information can influence how learners are understood by staff, so the prototype treats it as confidential support information.

The prototype follows these responsible-use principles:

- Data minimisation: no names, ID numbers, phone numbers, home addresses or unnecessary identifiers are required.
- Purpose limitation: the data is only for learner support planning, not discipline, exclusion, admissions or unrelated decisions.

- Transparency: the risk score is rule-based and shows reason codes so staff can understand why a learner was flagged.
- Fairness: access barriers such as limited internet or phone-only learning are treated as resource needs, not learner failure.
- Human review: risk flags are support signals only; a staff member must review context before any intervention.
- Harm prevention: the tool must not automate disciplinary, progression, funding or exclusion decisions.

Final interface controls also reinforce this approach:

- ethics confirmation before upload, manual entry, and export
- privacy mode in display settings
- responsible-use reminder messaging
- confirmation before upload and before clearing report state

The ethics review also includes a privacy notice, consent guidance, human-review workflow, data retention recommendation, bias risks, security recommendations and a governance checklist.

- Ethics review: docs/EthicsReview/ethics_review.md
- Ethics governance checklist:
docs/EthicsReview/ethics_governance_checklist.md

8. Methods, Tools, And Development Approach

Delivery followed a staged 15-day plan with iterative build-test-refine cycles and structured evidence capture. Later work focused on reliability hardening, settings architecture, UX refinements, and final QA evidence.

- Project plan: docs/FinalReport/project_plan.md
- 15-day plan: docs/FinalReport/15_day_delivery_plan.md

9. Testing, Bug Fixes, And Quality Evidence

Testing was completed across API behavior, validation paths, dashboard contract integrity, interaction flows, and frontend/backend build quality. Bugs were tracked and resolved with evidence for import parsing, risk explainability, upload feedback, modal behaviour, and health-check handling.

- Test log: evidence/Testing/test_log.md
- Bug log: evidence/BugLogs/bug_log.md

Final QA confirmed:

- Help modal opens and closes correctly
- Settings modal opens and API status checks work

- Ethics confirmation gates sensitive actions
- Every upload triggers a confirmation prompt
- New Report triggers a confirmation prompt before clearing state
- Frontend typecheck passes and backend Release build succeeds

10. Reflection

Personal learning note

This project helped me see that a dashboard is only useful if the data behind it is clean and the meaning of the numbers is clear. I also realised that risk scoring can be sensitive, so I kept the score explainable and included human review instead of presenting it as an automatic decision. A real version of this tool would still need review by programme staff and testing with properly anonymised data before being used operationally.

A major challenge was balancing the working prototype with all the evidence required by the assignment. Earlier versions were too focused on the dashboard and not enough on evidence mapping, testing, responsible-use wording, and reliability. I improved this by building around the assignment essentials first, then strengthening validation, ethics controls, settings behaviour, screenshots, and final evidence traceability.

11. Evidence Inventory Mapped To SFIA Skills

See docs/FinalReport/evidence_inventory_sfia.md.

12. Final Submission Checklist

- Working prototype: Complete
- Integrated report: Complete
- Synthetic dataset: Complete
- Data dictionary: Complete
- 5 visualizations: Complete
- 5 or more insights: Complete
- 3 or more recommendations: Complete
- As-is and to-be process models: Complete
- Business/value model: Complete
- Ethics and privacy review: Complete
- Ethics governance checklist: Complete
- Testing log and bug log: Complete

- README/user guide: Complete
- SFIA evidence checklist: Complete
- Completed peer/stakeholder review record: Complete
- Screenshot evidence regenerated: Complete
- Frontend typecheck verified: Complete
- Backend release build verified: Complete

Appendices

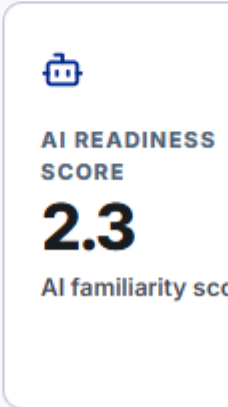
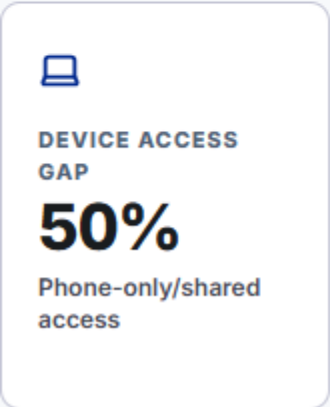
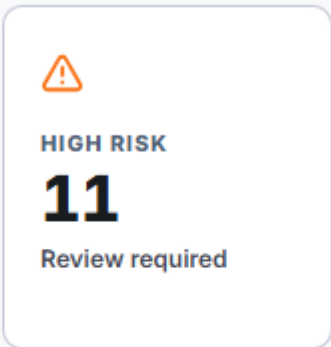
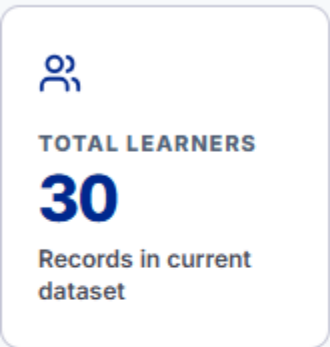
Evidence is packaged in the following order for submission review:

- README and user guide
- Dataset and data dictionary
- Process models and value model
- Ethics review, governance checklist, privacy notice and responsible-use evidence
- Testing log, bug log, and SFIA evidence checklist
- Peer or stakeholder review record
- Screenshot appendix and build evidence

Appendix A: Dashboard Overview

Shows the KPI cards and chart grid used for operational monitoring.

➤ Programme Support KPIs



Dashboard overview

Appendix B: High-Risk Learner Table

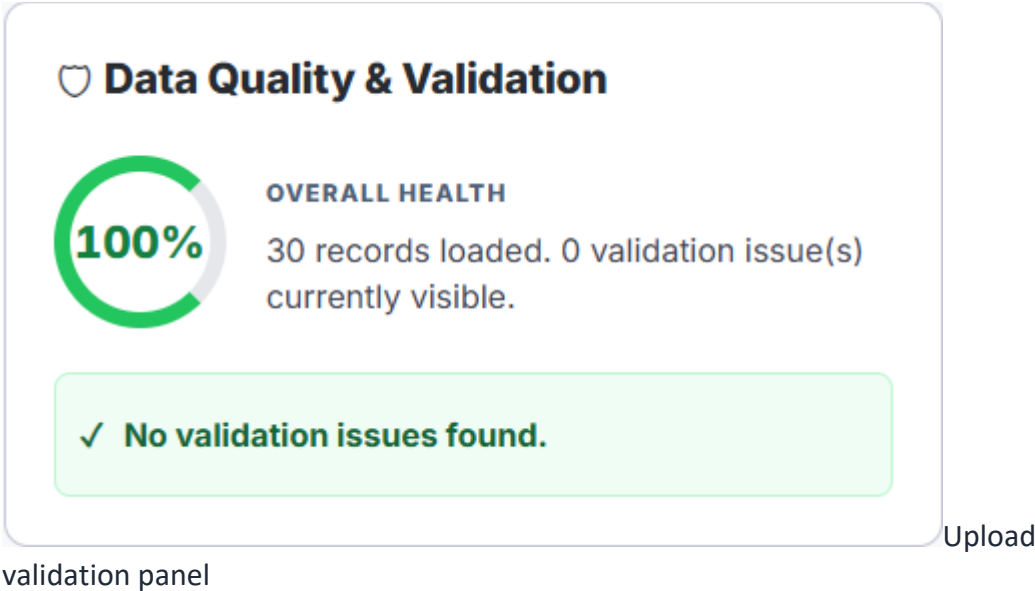
Shows the searchable risk table with score, priority, and factor chips for human-reviewed follow-up.

LEARNER ID	RISK LEVEL	KEY INDICATORS
02 L002 25-29	CRITICAL	Attendance High, Device Phone Only, Internet Unstable
06 L006 25-29	CRITICAL	Attendance High, Device Phone Only, Internet Unstable
09 L009 18-24	CRITICAL	Attendance High, Device Phone Only, Internet Limited
13 L013 18-24	CRITICAL	Attendance High, Device Phone Only, Internet Limited
17 L017 18-24	CRITICAL	Attendance High, Device Phone Only, Internet Limited
21 L021 18-24	CRITICAL	Attendance High, Device Phone Only, Internet Limited
25 L025 18-24	CRITICAL	Attendance High, Device Phone Only, Internet Limited
29 L029 18-24	CRITICAL	Attendance High, Device Phone Only, Internet Limited
05 L005 18-24	CRITICAL	Attendance High, Internet Limited, Digital Confidence Low
15 L015 18-24	HIGH	Attendance Medium, Device Shared Laptop, Internet Unstable

High-risk learner table

Appendix C: CSV Upload Validation

Shows the row-level validation panel produced when invalid CSV values are uploaded.



Appendix D: Manual Form Validation

Shows inline field-level validation on the manual learner entry form.

Manual Learner Entry

L

Learner ID must be at least 3 characters.

18-24

W

Province is required.

Laptop

Reliable

0

Use a whole number from 1 to 5.

0

Use a whole number from 1 to 5.

0

Use a whole number from 1 to 5.

Employment status

Employment status is required.

Support need

Manual form validation

Appendix E: Recommendations And Ethics

Shows the recommendations panel alongside the ethics and privacy panel.

Programme Recommendations

Provide connectivity and device support

15 learners report limited or unstable internet access. Meaningful share rely on phone-only or slow access.

View Action →

Launch targeted attendance support

11 learners match high-risk conditions (high risk and confidence/access constraints).

View Action →

Introduce confidence-boost learning

17 learners show low average confidence in digital/programming/AI skills.

View Action →

Responsible Use Reminder

Risk scores and recommendations are designed to identify 'potential problems'. Staff remain responsible for identifying if 'this is a problem'.

✓ Consent considered

✓ Data anonymised where possible

+ New Report

Recommendations and ethics panels

Appendix F: Frontend Build Evidence

Successful frontend production build output captured for submission evidence.

Frontend Build Output

```
> student-support-insights-frontend@1.0.0 build
> vite build
```

```
vite v5.4.21 building for production...
transforming...
```

```
889 modules transformed.
```

```
rendering chunks...
```

```
computing gzip size...
```

dist/index.html	0.59 kB
dist/assets/index-DQoLVBBw.css	9.67 kB
dist/assets/DashboardCharts-D7mINDEj.js	5.17 kB
dist/assets/LearnerForm-nzVA0ahA.js	5.52 kB
dist/assets/index-1sM-toP3.js	19.58 kB
dist/assets/http-vendor-DcNlVx-A.js	42.33 kB
dist/assets/charts-vendor-zIFTg3jN.js	264.25 kB
dist/assets/vendor-BojNP3JM.js	284.52 kB

```
built in 3.83s
```

Frontend build evidence

Appendix G: Backend Build Evidence

Successful backend release build output captured for submission evidence.

Backend Release Build Output

Determining projects to restore...

Restored c:\Users\Yonela\New folder (3)\student-support-insights-tool\backend

StudentSupportInsights -> c:\Users\Yonela\New folder (3)\student-support-insi

Build succeeded.

0 Warning(s)

0 Error(s)

Time Elapsed 00:00:10.08

Backend build evidence

Final Note

This submission document is aligned to the current state of the regenerated integrated report in docs/FinalReport/integrated_report.md, but keeps the structure concise for assessor-facing packaging.